

PQC-Readiness Audit Report

Contract: /mnt/d/Crypto/audit_case_studies/contracts/ENS_NameWrapper.sol
SHA-256: 2c791aabd5c05746ce473cbe07d8c136... **Lines:** 1180 **Date:** 2026-08-20 **Tool:** pqc-audit v0.2.0

Executive Summary

Check	Result	Detail
Security Scan	WARN	0 critical, 1 high
PQC Readiness	WARN	0 quantum-vulnerable primitives, 0 critical implementation advisories
Gas Optimization	95 opportunities	~19000 gas units
OVERALL	PASS WITH WARNINGS	

Phase 1: Structure

- **Solidity Version:** ~0.8.17
- **Contracts:** 6
- **Functions:** 51
- **Events:** 0

Phase 2: Gas Optimization (95 found)

#	Pattern	Impact	Fixable
1	Use uint256 instead of smaller uints	Info	Manual
2	Use uint256 instead of smaller uints	Info	Manual
3	Use uint256 instead of smaller uints	Info	Manual
4	Use uint256 instead of smaller uints	Info	Manual
5	Use uint256 instead of smaller uints	Info	Manual

Phase 3: Security Analysis (42 findings)

Severity	Count
Critical	0
High	1
Medium	23
Low	18

Security Score: 0.0/100

Top Findings

- **[MEDIUM]** Line 124 — Missing zero address check
 - Code: `if (owner == address(0)) {`
 - Missing zero address check for critical operations.
- **[MEDIUM]** Line 208 — Missing zero address check
 - Code: `if (address(upgradeContract) != address(0)) {`
 - Missing zero address check for critical operations.
- **[MEDIUM]** Line 215 — Missing zero address check
 - Code: `if (address(upgradeContract) != address(0)) {`
 - Missing zero address check for critical operations.
- **[MEDIUM]** Line 362 — Missing zero address check
 - Code: `registrarOwner != address(this) ||`
 - Missing zero address check for critical operations.
- **[MEDIUM]** Line 363 — Missing zero address check
 - Code: `ens.owner(node) != address(this)`
 - Missing zero address check for critical operations.

Phase 4: Post-Quantum Cryptography Readiness

Line	Primitive	Risk	Migration Path
285	keccak256	MEDIUM	Double hash output length or use SHA-3-512...

Quantum Risk Level: HIGH

NIST Status: FIPS 204 (ML-DSA) and FIPS 205 (SLH-DSA) finalized. FIPS 206 (FN-DSA/Falcon) selected for standardization; publication pending.

Implementation Vulnerability Scan (2024-2026 Research)

Vulnerability	Risk	Source
<i>None detected</i>	LOW	No known implementation vulnerabilities match this contract

Evidence Attestation

- **Source SHA-256:** 2c791aabd5c05746ce473cbe07d8c1364ac5ed69c495caa5bea79536f0363c80
- **Report SHA-256:** 4bca8c6e75a8328fcf21552baad0075d...

This report was generated mechanically. No LLM inference was used in the security or PQC assessments. PQC-Readiness Audit — pqc-audit v0.2.0